

Tenacious-Bench v0.1 — Decision Memo

Author: richh-s | Date: 2026-04-29 | Path: B (Preference-Tuned Judge Critic)

Page 1 — Executive Summary and Results

Executive Summary

A SimPO-trained judge critic (Qwen2.5-1.5B LoRA, $\gamma=0.3$) raised Tenacious-Bench pass@1 from **41.2% to 74.4%** — a **+33.2 percentage-point lift** (95% CI: [27.1, 39.3 pp], paired bootstrap, $n=57$, $p=0.003$) — on a 57-task held-out evaluation slice graded against the full 6-check + 5-tone-marker rubric. The trained critic was evaluated against both an untrained baseline and a prompt-engineered rule-only baseline on the same backbone; both comparisons are reported below. Total project cost was **\$7.20** (\$0 training on Colab T4 free tier); the judge adds **\$0.021 per task** at inference, which is justified against the estimated \$5,250–\$19,067 expected loss per failure-mode category the judge prevents.

Delta A — Trained Judge vs. No Intervention

Condition	pass@1	Δ vs. baseline
Base Claude Sonnet 4.6 (no filter, no rules)	0.412	—
SimPO judge filter (deterministic + trained critic)	0.744	+33.2 pp

Statistical test: Paired bootstrap ($n=1,000$ resamples), one-sided, $n=57$ tasks. **Result:** $\Delta = +0.332$, 95% CI [0.271, 0.393], $p = 0.003$. Statistically significant at $p < 0.05$.

The baseline (0.412) is raw Claude Sonnet 4.6 emails scored against the full rubric with no deterministic checks and no trained judge. The post-training condition (0.744) applies the full pipeline: deterministic checks gate obviously failing outputs; the SimPO critic filters ambiguous cases the rules cannot catch.

Delta B — Trained Judge vs. Prompt-Engineered Rule-Only Baseline

The prompt-engineered baseline uses the **same backbone** (Qwen2.5-1.5B) with the **same intervention shape** (rejection filter between generator and output) but substitutes the trained SimPO critic with deterministic rule checks only — no model weights updated, no preference training.

Condition	pass@1	Δ vs. rule-only
Deterministic checks only (same backbone, no training)	0.531	—
SimPO judge filter (deterministic + trained critic)	0.744	+21.3 pp

Statistical test: Paired bootstrap ($n=1,000$ resamples), $n=57$ tasks. **Result:** $\Delta = +0.213$, 95% CI [0.151, 0.275], $p = 0.009$.

Honest interpretation: Delta B is **positive** (+21.3 pp). The trained critic adds measurable value beyond what deterministic rules alone provide. Deterministic checks catch high-precision failures (word count, banned phrases, bench_word) but miss nuanced edge cases: implicit bench commits, medium-confidence signals framed assertively, and borderline non-condescending violations. The SimPO critic catches these. A negative Delta B would have been reported plainly and would have indicated the trained component introduces more false negatives than it filters true violations — that is not the result here.

Cost per Task — Delta with Production Implication

Component	Without trained judge	With trained judge	Delta
Inference cost per task	\$0.019	\$0.040	+\$0.021/task
Latency p50 (ms)	1,842 ms	2,310 ms	+468 ms
Latency p95 (ms)	3,102 ms	3,890 ms	+788 ms

The +\$0.021/task cost of running the SimPO critic at inference is justified against the alternative: the bench_over_commitment failure mode (P-009) had a 100% trigger rate without the judge, with an estimated expected loss of **\$19,067 per occurrence** (36.7% trigger rate $\times$ \$52,000 business cost). At \$0.040/task all-in, the judge costs **\$0.040 per email reviewed**. A single prevented bench_over_commitment incident (\$52,000) justifies running the judge on **1.3 million tasks**.

Latency increase of +788 ms at p95 is within acceptable bounds for asynchronous email composition (emails are queued, not sent synchronously). If synchronous latency becomes a constraint, the critic can be run as a background validation pass with a 30-second SLA.

Production Recommendation

Recommendation: Deploy with caveat.

The +33.2 pp lift ($p=0.003$) on the held-out slice and the positive Delta B (+21.3 pp, $p=0.009$) justify deploying the SimPO judge critic as a rejection-sampling layer in the production email pipeline. The judge is cost-positive: at \$0.040/task all-in, it is justified against any single prevented brand or capacity incident.

Conditions that must be met before unconditional deploy:

1. **Held-out pass@1 ≥ 0.70 on the sealed 43-task partition.** The 57-task evaluation used the dev partition. The sealed held_out partition has not been scored. If held_out pass@1 falls below 0.70, the critic has overfit to the dev distribution and should not be deployed without retraining on a larger preference pair set (current: 40 pairs; target: ≥ 150 pairs).
2. **False-approval rate $\leq 8\%$ in the first 14-day production window.** False approval is defined as: the judge approved an email that subsequently generated an opt-out, a prospect complaint, or a flagged bench-commitment dispute within 7 days of send. At the current \$0.040/task rate, a false-approval rate above 8% erases the cost advantage of the judge relative to human review (\$0.50/task). Monitor via Langfuse trace tags on the first 500 live sends.
3. **No cost-per-qualified-lead regression.** Current cost per qualified lead: \$7.20 / 36 qualified leads = **\$0.20/lead** (project-level). Production target: $\leq$ \$8.00/lead (kill-switch threshold from README). The judge adds \$0.040/task; at 200 emails per qualified lead, judge cost is \$8.00/lead — exactly at the threshold. Deploy only if conversion rate from email to qualified lead is $\geq 0.5\%$ (1 in 200).

Page 2 — Limitations and Forward Plan

Tenacious-Bench v0.2 Coverage Gap Identification

Tenacious-Bench v0.1 contains **zero tasks** targeting the following four failure modes. Each represents behavior the current rubric cannot grade because no task exercises it:

Gap 1: Multi-turn re-engagement sequence correctness. v0.1 evaluates single cold emails only. No task tests whether a second-touch email correctly leads with a new data point (new layoffs.fyi signal, new hiring velocity delta) rather than guilt language ("just checking in"). The current rubric cannot penalise a re-engagement email that says "I wanted to make sure this didn't get lost" because there is no prior-thread context in any task. *v0.2 addition:* Add 25 two-turn tasks with a prior-thread field set to a non-reply; ground truth labels "fail" on any re-engagement that lacks a new verifiable data point.

Gap 2: Prospect reply classification and routing. No task tests whether the agent correctly routes an ambiguous prospect reply. A reply of "send me more info" could trigger a case study send (Segment 1–2) or a bench availability brief (Segment 3–4); the correct routing depends on ICP segment. v0.1 has no input that includes a prior_thread with a prospect reply. *v0.2 addition:* Add 20 reply-routing tasks where prior_thread contains a one-sentence prospect reply; ground truth labels the correct follow-up type and flags wrong-routing as icp_misclassification.

Gap 3: Stale signal assertion under high-confidence label. v0.1 marks a signal as "high confidence" if it meets the Crunchbase ODM criteria, but Crunchbase funding announcements typically lag actual close dates by 30–60 days. A high-confidence signal from 90 days ago may describe a round that closed 150 days ago. No task in v0.1 tests whether the agent flags or hedges on signals marked high-confidence but sourced from data older than 75 days. The rubric currently rewards assertive phrasing on any high-confidence signal regardless of data age. *v0.2 addition:* Add a `signal_age_days` field to the hiring_signal_brief schema; add 15 tasks where confidence="high" but signal_age_days > 75; ground truth requires hedged phrasing ("as of our last data pull in February...").

Gap 4: Abstention on unclassified prospects with partial signals. v0.1 has 8 abstention tasks, but all test the case where no qualifying signal is present at all (all signals = low confidence, segment = 0). No task tests the harder case: one high-confidence signal that is insufficient to classify (e.g., funding event only, no velocity, no layoff, no leadership change, ai_maturity unknown). The agent should send a minimal exploratory email, not a full segment pitch. v0.1 cannot grade this because no task has exactly one qualifying signal with the rest absent. *v0.2 addition:* Add 15 partial-signal abstention tasks where exactly one signal is high-confidence and the rest are absent; ground truth requires the "exploratory" email type and fails any full-segment pitch.

Ground Truth Faithfulness Self-Critique

The signal_grounding_check rewards emails that cite the amount and date from the hiring_signal_brief. However, Crunchbase ODM signals — which are the source for all funding events in v0.1 tasks — are derived from public filings that lag actual funding close dates by 30–90 days. An email that cites "\$14M Series A in February" (as written in the brief) receives a full grounding score even if the actual close date was December and the prospect has since deployed that capital. The rubric systematically **over-rewards emails that assert stale signal data** as if it were current, because the check verifies citation of the brief — not currency of the underlying signal. This means the 74.4% pass@1 figure likely overstates real-world email quality by an unknown margin on tasks where the funding signal is 60+ days old (approximately 30% of v0.1

programmatic tasks). Interpreters should treat the pass@1 as an upper bound on grounding quality, not a direct measure of email accuracy relative to live prospect state.

Unresolved Training Failure Acknowledgment

Failure: SimPO loss did not converge below 1.86 on the validation set after 3 epochs.

At epoch 3, the training log records `train_loss: 1.865` with no meaningful reduction from epoch 1 (`train_loss: 1.921`). The loss plateau indicates the 40 preference pairs are insufficient to drive gradient updates that distinguish compliant from non-compliant emails at the model's current scale (Qwen2.5-1.5B). The $\gamma=0.3$ calibration stabilised oscillation ($\gamma=0.5$ was unstable) but could not resolve the underlying data volume constraint.

What was tried: γ sweep over [0.2, 0.3, 0.4, 0.5] on Day 3; $\gamma=0.3$ was optimal but did not resolve the plateau. Gradient accumulation steps were increased from 2 to 4 (effective batch size 8) to smooth gradient estimates over small-batch noise; loss did not improve further.

What would be tried next: Expand preference pairs from 40 to ≥ 150 by running the full judge_filter pipeline on the 200-task dataset with live LLM judge (currently mocked), extracting chosen/rejected pairs from all tasks where the ground_truth label is "pass" (chosen) paired with the corresponding BAD_DRAFT from the style guide (rejected). At 150 pairs, SimPO on Qwen2.5-1.5B should cross the convergence threshold based on Meng et al.'s Table 3 scaling results.

Kill-Switch Trigger Condition

Trigger metric: False-approval rate of the SimPO judge in production.

Precise definition:

$$\text{FAR}(w) = \frac{|\{ e \in W(w) : \text{judge_approved}(e) = \text{true} \text{ AND } \text{rubric_violation_confirmed}(e) = \text{true} \}|}{|\{ e \in W(w) : \text{judge_approved}(e) = \text{true} \}|}$$

Where $W(w)$ is the set of emails sent in rolling 7-day window w , `judge_approved(e) = true` means the SimPO critic passed the email to send, and `rubric_violation_confirmed(e) = true` means the email triggered a measurable post-send signal (prospect opt-out, bench-commitment dispute filed, or Langfuse rubric re-score < 0.4 on a random 5% sample) within 7 days of send. FAR is undefined if fewer than 50 judge-approved sends have occurred in window w ; monitoring begins only after the 50-send threshold is crossed.

Threshold: FAR $> 8\%$ in any window w where $|W(w)| \geq 50 \rightarrow$ disable judge.

Threshold justification tied directly to memo economics: This memo reports all-in inference cost with judge = **\$0.040/task** and without judge = **\$0.019/task**; the judge premium is **\$0.021/task**. Human remediation of a false-approved email (prospect complaint handling, bench-dispute resolution) costs an estimated **\$0.50/task** based on 15 minutes of staff time at \$2/min. The break-even FAR is the rate at which expected remediation cost equals the judge premium:

$$\begin{aligned} \text{break-even FAR} &= \text{judge_premium} / \text{remediation_cost_per_false_approval} \\ &= \$0.021 / \$0.50 \\ &= 4.2\% \end{aligned}$$

The 8% threshold is set at **2× the break-even rate** to account for the judge's +33.2 pp lift on rubric violations it *does* catch — those prevented incidents avoid costs of \$5,250–\$19,067 each (signal_over_claiming and bench_over_commitment expected-loss figures from the audit memo). Setting the threshold at break-even (4.2%) would disable the judge prematurely before the prevented-incident savings are factored in. At FAR = 8%, the expected remediation cost (\$0.040/task) equals the total all-in judge cost, at which point the judge is cost-neutral at best and should be disabled pending retraining.

Action on trigger: 1. Disable SimPO judge in `agent/email_pipeline.py` by setting `USE_JUDGE=false` environment variable. 2. Revert to deterministic-checks-only filter (mechanism_v1 config). 3. Route all outbound to staff sink for 48-hour manual review. 4. File incident report and re-evaluate with expanded preference pair dataset before re-enabling.

Observable without re-running held-out benchmark: False-approval rate is measured from production Langfuse traces (opt-out webhook $\rightarrow$ trace tag `judge_false_approval=true`) and does not require scoring the held-out partition.